

Ishaan Shrivastava

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Career Summary

My current objective is to make vision-language models learn how to reason better for less.
Machine-Learning Engineer, published paper on multimodal vision-language models (“Learn ‘No’ to Say ‘Yes’ Better”) cited 30 times, rated among the top 7% of competitive programmers on codeforces, ranked number 1 on the tensortonic leaderboard. Working on multimodal VLMs and shipping computer vision solutions on a variety of tasks at Metafusion.

Skills and Technologies

Programming Languages: Python (4+ years), C++/C (4+years), MATLAB, Git
Open-source libraries: PyTorch, Jax, einops, Numpy, OpenCV, Pillow

Coursework

AI and Data Science

- Pattern Recognition and Machine Learning
- Data Engineering
- Artificial Intelligence
- Advanced Artificial Intelligence
- Deep Learning
- Dependable AI

Mathematics

- Probability, Statistics and Stochastic Processes
- Mathematics I & II
- Optimization for Machine Learning
- Introduction to Game Theory
- Introduction to Financial Engineering

Computer Science

- Intro to Computer Science
- Data Structures and Algorithms
- Principles of Computer Systems I & II (Operating Systems and Networking)

Publications

Learn “No” to Say “Yes” Better: Improving Vision-Language Models via Negations - *arXiv preprint*
Jaisidh Singh, Ishaan Shrivastava*, Mayank Vatsa, Richa Singh, Aparna Bharati* Published - *WACV2025*

Work Experience

Computer Vision Research Engineer

Metafusion [Supervisor: Manish Soni]

May '24 – Current

Noida, India

- Curated 20+ benchmarks for object detection and performed ablations, boosting narrow-domain performance over SOTA by 2% and yielding an improved pretrained object detection base for downstream finetuning.
- Independently discovered a **data-free structured pruning** method for attention-layer skipping in VLMs by leveraging insights from **heavy-tailed self-regularization theory (HTSR)**
- Research and development of a person-attribute recognition pipeline resistant to distribution shifts. Successfully deployed across many different applications.

- Optimisation and backend fixes for **Multi Object Tracker (MOT)**, leading to an average of 12-15% performance increase over a wide range of MOT metrics.
- Development and maintenance of **FuseNets**, a deep-learning library created to accelerate and automate the R&D and deployment of novel computer vision model architectures on detection, segmentation, and recognition.
- Responsible for training and deploying a variety of **I3d feature**-based approaches for person attribution and violence detection on low quality video surveillance feeds in low data conditions.

Open source engineer

May '23 – Aug '23

CodeForGovTech '23 [Supervisor: Gautam Rajeev]

Remote

- Devised and implemented a low-cost framework for improving translations in transformer-based models using a dictionary of translations.
- Improved google's **Tesseract OCR** performance on multilingual document parsing using **OpenAI's CLIP** and published multiple open source libraries for further development and refinement around the same.
- Worked with supervisors for **successful deployment** of multilingual parser, positively impacting **millions of farmers in underrepresented** and underprivileged language communities.

Undergraduate researcher

May '23 – Aug '23

Supervisor: Dr. Shweta Jain

Ropar, Punjab, India

- Worked on improving fairness in non-i.i.d federated classification settings using knowledge distillation, EMA, and stochastic weight averaging.
- Implemented the **Ditto** framework for federated learning in PyTorch and ran experiments on it.
- Achieved an improvement of 0.2% upon other baselines.

AI researcher

Dec '22 – Feb '23

MURVEN [Supervisor: Murugan Venugopal]

Chennai, Tamil Nadu, India

- Analysis, design, coding, integration, debugging and deployment of diffusion models using AI to prompt the generation of video animations
- Working on a reusable and intuitive framework for prompting **Stable Diffusion** to generate animations

Computer Vision Engineer Intern

Nov '22 – Feb '23

QwizPad [Supervisor: Mohit Shrivastava]

Chandigarh, Punjab, India

- Integrated a digit recognition model into existing digital evaluation framework in .NET using inter-process communication
- eliminated disk read-writes to enable very high speed scanning and inference of OMR sheets

Undergraduate researcher

Jan '22 – Jul '22

Supervisor: Dr. Angshuman Paul

Jodhpur, Rajasthan, India

- Worked on highly imbalanced fine grained spine and chest x-ray image classification on **VinDr-SpineXR** and **VinDr-PCXR** datasets
- Developed a general - purpose class-incremental learning pipeline with **Knowledge Distillation** and **DenseNet**
- Set a baseline for further class incremental learning research in this area

Projects

DevRev Expert Answers in a Flash:Improving Domain-Specific QA - [Colab](#) Dec '23 – Jan '23

Supervisor: Self Project

- Implemented a Document retrieval and extractive question answering pipeline using pre-trained LLMs in Low-resource settings without GPU.
- Reduced overall pipeline inference time (document retrieval from an corpus of data + extractive QA) to under 700 ms per query.
- Conducted an extensive ablation study of approached used in order to push the inference time and evaluation metrics.

Self-Supervised robust Adversarial Attack Detection - [Github](#) Jan '23 – Jan '23

Supervisor: Prof. Richa Singh, Prof. Mayank Vatsa, CSL7370 (Dependable AI)

- Implemented an Autoencoder to maximise the reconstruction error of adversarially attacked samples by implementing a self-supervised triplet loss objective by mimicing the behaviour of attacked samples.
- Reproduced results on the original datasets as well as new ones mentioned in the original paper.

FRUITS-360 classifier web-app - [Github](#) March '22 – May '22

Supervisor: Prof. Richa Singh, CSL2050 (Pattern Recognition and Machine Learning)

- Explored a wide variety of ML algorithms - Decision trees, Random Forest, XGBoost / LightGBM, SVMs, MLP
- Made a pipeline for classifying fruit images trained on Fruits-360 data.
- Hosted a webpage with a fine-tuned VGG16-based model as backend

Flight price prediction March '22 – April '22

Supervisor: Prof. Richa Singh, CSL2050 (Pattern Recognition and Machine Learning)

- Used the google API to gather Google Trends data on festivals to improve flight price prediction
- Tested various machine learning approaches - SFS, decision tree regressors, etc.
- Made and deployed a pipeline of ensemble models trained on the provided dataset.

Education

Indian Institiue of Technology, Jodhpur (present) November '20 – May '24

B.Tech in Artificial Intelligence and Data Science - current CGPA 8.1 / 10

Jodhpur, Rajasthan, India

Stepping Stones Sr. Secondary School

March '18 – Feb '20

Central Board of Secondary Education - CGPA 8.96 / 10

Chandigarh, Punjab, India

Strawberry Fields High School

March '07 – Feb '18

Council For The Indian School Certificate Examinations - CGPA 8.3 / 10

Chandigarh, Punjab, India

Positions of Responsibility

Core Member - Programming Club, IIT Jodhpur

August '22 – May '24

- Problemsetter for Programming Club's monthly competitive programming contest, Dinnertime, and IIT Jodhpur's Tech Fest, Prometeo
- Inculcated programming culture and interest for competitive programming among the community

Achievements

Expert at Codeforces Reached 'Expert' (rating 1749, 7 months, 33 contests) – top 7 % on Codeforces

Other Interests

Avid chess player, origami enthusiast, and active competitive programmer (Codeforces 'Expert'). I thrive under competition.